

ReefSafe Data Provenance & Dataset Directory

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Executive Architecture Summary: ReefSafe implements an institutional, transparent data pipeline combining 19 years of verified longitudinal coral reef survey archives (2007–2025) and a 14-year predictive modeling panel (2012–2025: 6,381 site-year rows across 560 sites and 56 islands) with national macroeconomic series and satellite thermal diagnostics. To maintain complete scientific integrity, **all datasets are catalogued with explicit usage boundaries:** only verified ecological transects enter the scored Gradient Boosting ML model, while external satellite signals (NOAA DHW), macroeconomic indicators (DOSM Real GDP), and historical benchmarks (DMPM TEV) serve as descriptive context. Below is the complete catalogue of all underlying datasets, their publishers, and direct download links.

1 Primary Ecological Monitoring Panel (Scored Model Input)

Reef Check Malaysia (RCM) Annual Survey Reports (2007–2025)

VERIFIED INPUT

SCORED ML

Standardized 100m transect surveys across 560 registered monitoring sites on 56 islands/units in national marine parks and coastal reefs, yielding an unbalanced panel of 6,381 site-year observations (2012–2025) with 5,821 valid temporal transitions (and 444 paired 2024–2025 comparison sites). Supplies core inputs to the Gradient Boosting Regressor: Live Coral Cover (LCC %), substrate categories (hard coral, soft coral, dead coral, algae), disturbance impact mentions (bleaching, warm water, storm, sedimentation), and indicator fish/invertebrate counts.

Annual Reports Archive: <https://reefcheck.org.my/annualsurveyreports/> [DOWNLOAD PORTAL](#)

2024 Bleaching Report: <https://reefcheck.org.my/wp-content/uploads/2025/07/2024CoralBleachingImpactReportMalaysia.pdf> [DIRECT PDF](#)

Official Homepage: <https://reefcheck.org.my>

Datathon Drive Archive: Google Drive → "Datathon Data Archive" → Subfolder "RC Annual Reports" (19 raw survey PDFs)

2 Department of Statistics Malaysia (DOSM) & data.gov.my

State Real GDP by Economic Activity (Supply Side)

VERIFIED CONTEXT

OPEN DATA

Official state-level real GDP supply series covering Accommodation, Food & Beverage, Transportation, and Services. Used to evaluate coastal tourism dependencies and maritime economic exposure across Peninsular Malaysia, Sabah, and Sarawak.

Dataset Landing & Download: https://data.gov.my/data-catalogue/gdp_state_real_supply [CATALOGUE](#)

Direct API Query (JSON): https://api.data.gov.my/data-catalogue?id=gdp_state_real_supply [DIRECT DATA](#)

OpenDOSM GDP Portal: <https://open.dosm.gov.my/dashboard/gdp>

Marine Fish Landings by State (Resource Pressure Proxy)

VERIFIED CONTEXT

OPEN DATA

Published by Department of Fisheries Malaysia (DOFM) via OpenDOSM / data.gov.my. Annual marine fish landings (tonnes) disaggregated by state and coast (East Coast, West Coast, Sabah, Sarawak). Serves as regional proxy for maritime extractive resource pressure.

Dataset Landing & Download: https://data.gov.my/data-catalogue/fish_landings [CATALOGUE](#)

Direct API Query (JSON): https://api.data.gov.my/data-catalogue?id=fish_landings [DIRECT DATA](#)

River Basin Water Quality & Pollution Indicators (Runoff Proxy)

VERIFIED CONTEXT

OPEN DATA

Published by Department of Environment (DOE) via data.gov.my. Monitored river basin counts and water quality classifications (Biochemical Oxygen Demand / BOD, Ammoniacal Nitrogen, Suspended Solids). Provides descriptive context on coastal catchment runoff.

Dataset Landing & Download: https://data.gov.my/data-catalogue/water_pollution_basin [CATALOGUE](#)

Direct API Query (JSON): https://api.data.gov.my/data-catalogue?id=water_pollution_basin [DIRECT DATA](#)

Domestic Tourism Survey 2024 (Tourism Spending Metric)

VERIFIED CONTEXT

NATIONAL STAT

Published by DOSM. Establishes the authoritative national average expenditure per domestic visitor (RM410), directly utilized in the Reef-Adjacent Economy proxy formula: (Visitors) × (RM410 Spend) × (10% Reef Share).

Official Survey Release: <https://www.dosm.gov.my/portal-main/release-content/domestic-tourism-survey-2024>

OFFICIAL RELEASE

3 Historical Marine Park Visitors Series 2000–2017 (archive.data.gov.my)

Department of Marine Park Malaysia Visitor Records (2000–2017)

VERIFIED CONTEXT

DIRECT CSV

Official 18-year historical visitor series across 5 maritime jurisdictions. All 90 rows transcribed in `taman_laut_visitors_2000_2017.csv` were verified against published government records (Creative Commons Attribution). The 2013–2017 average (1,464,770 visitors across Payar, Tioman, Redang, Tinggi) serves as the national baseline for the Reef-Adjacent Economy calculation.

Johor Marine Park CSV: [Direct CSV Download \(Johor\)](#) [EXACT CSV](#)

Kedah Marine Park CSV: [Direct CSV Download \(Kedah\)](#) [EXACT CSV](#)

Pahang Marine Park CSV: [Direct CSV Download \(Pahang\)](#) [EXACT CSV](#)

Terengganu Marine Park CSV: [Direct CSV Download \(Terengganu\)](#) [EXACT CSV](#)

Labuan Marine Park CSV: [Direct CSV Download \(Labuan\)](#) [EXACT CSV](#)

Web Dataset Landing: archive.data.gov.my/data/ms_MY/dataset (Archive Web Catalogue)

4 NOAA Coral Reef Watch Satellite Time-Series (Regional Heat Context)

5km Satellite Virtual Stations Daily SST & Degree Heating Weeks (DHW)

ELIGIBILITY PENDING

CONTEXT ONLY

Published by NOAA Coral Reef Watch (U.S. NESDIS / NOAA). Provides operational daily satellite sea surface temperature (SST), SST anomaly, bleaching hotspots, and Degree Heating Weeks (DHW) accumulation. **Strict Boundary:** Kept un-scored outside the ML model; serves purely as regional contextual signal for rangers to correlate observed bleaching events with regional heat stress.

Time-Series Download Directory: <https://coralreefwatch.noaa.gov/product/vs/data.php> [DATA DIRECTORY](#)

Interactive Virtual Stations Map: <https://coralreefwatch.noaa.gov/product/vs/map.php> [INTERACTIVE MAP](#)

Virtual Stations System Info: <https://coralreefwatch.noaa.gov/product/vs/index.php>

NOAA CRW Main Portal: <https://coralreefwatch.noaa.gov>

5 Geospatial & Marine Park Administrative Bodies

JUPEM Geodetic Baselines & DOFM Marine Park Registers

VERIFIED CONTEXT

ADMINISTRATIVE

Authoritative spatial references defining island coordinates, maritime baselines, gazetted Marine Park boundaries, and Fishery Prohibited Areas (FPAs).

JUPEM Geodetic Portal: <https://www.jupem.gov.my> [OFFICIAL PORTAL](#)

DOFM Marine Parks Section: <https://www.dof.gov.my> [OFFICIAL REGISTER](#)

6 Economic Valuation Benchmarks & Supporting Methods

Total Economic Value (TEV) of Marine Biodiversity & Tourism Attribution Method

VERIFIED CONTEXT

METHODOLOGY

Historical economic valuation benchmarks and peer-reviewed international methodologies for coastal reef economic attribution:

DMPM TEV Benchmark Report: https://wdpa.s3.amazonaws.com/Country_informations/MYS/TOTAL%20ECONOMIC%20VALUE%20OF%20MARINE%20BIODIVERSITY.pdf [DIRECT PDF](#)

Spalding et al. (2017) Method: https://www.nature.org/content/dam/tnc/nature/en/documents/paper_coralreeftourism_spalding_2017.pdf [DIRECT PDF](#)

Sabah Parks 2024 Dashboard: <https://dashboard.sabahparks.org.my>

Terengganu 2024 Arrivals Release: <https://malaysia.news.yahoo.com/tourist-arrivals-rise-terengganu-islands-050917502.html>

7 Provenance Register & Boundary Governance Matrix

Machine-readable source register from data/provenance_manifest.csv, establishing admissible scopes and interpretation guardrails.

Dataset ID & Title	Publisher	Temporal / Spatial Coverage	Status	Platform Role	Governing Limitation / Boundary
reef_check_surveys Annual Reef Survey Reports	Reef Check Malaysia (RCM)	2012–2025 (Archive 2007–2025) 6,381 site-year rows (560 sites, 56 islands)	VERIFIED_INPUT	Primary Scored ML Input (Live Coral Cover %)	Unbalanced panel of 560 registered sites across 56 islands (5,821 valid temporal transitions; 444 paired 2024–2025 sites); narrative flags record mentions rather than confirmed absence.
dmpm_tev_2011_2015 Total Economic Value (TEV)	Department of Marine Park Malaysia	2011–2015 studies 6 archipelagos	VERIFIED_CONTEXT	Historical Economic Valuation Benchmark	RM8.7B is a rounded annual valuation benchmark across 6 archipelagos; not ReefSafe revenue, asset value, or avoided loss.
noaa_crw_virtual_stations 5km Satellite Virtual Stations	NOAA Coral Reef Watch (U.S. NESDIS)	Daily time-series 5 regional stations	ELIGIBILITY_PENDING	Descriptive Heat Stress Context Only	Kept un-scored outside the ML model; satellite proxies rather than in-water logger measurements.
marine_park_visitors_2000_2017 State Marine Park Visitors	DMPM / data.gov.my archive	2000–2017 90 state-year rows	VERIFIED_CONTEXT	Visitor Volume Baseline (2013–2017 avg: 1.46M)	State-level aggregates cannot be assigned to individual monitoring units or commercial island operators.
dosm_domestic_tourism_2024 Domestic Tourism Survey	Department of Statistics Malaysia	2024 Annual Release	VERIFIED_CONTEXT	RM410 Spending Coefficient	National domestic expenditure average; serves as a spending proxy, not a commercial turnover forecast.
island_arrivals_sabah_2024 Sabah Parks Statistics	Sabah Parks	2024 Full Year 5 park jurisdictions	VERIFIED_CONTEXT	Park-level Context for Reef-Adjacent Estimates	Park totals with domestic/international split; parks span multi-island archipelagos.
bleaching_impact_2024 4th Global Bleaching Event	Coralku & RCM (Szere day 2025)	2024 Bleaching Event 26 site records	VERIFIED_CONTEXT	Bleaching Mortality Context	Headline 34.1% average mortality is the published report average, not a model re-estimation.
opendosm_economic_series State GDP, Fish, Water Quality	DOSM / data.gov.my	Annual State Series	VERIFIED_CONTEXT	Descriptive Macroeconomic & Catchment Context	State-level indicators do not measure monitoring-unit tourism pressure or direct causality.
island_accommodations Accommodation Inventory	Unknown Local Compilation	Static 56-island table	EXCLUDED_UNVERIFIED	EXCLUDED from Platform	No reproducible source audit trail supports completeness; excluded from all features, rankings, and claims.

8 Methodological Guardrails & Concept Disambiguation

Reef-Adjacent Economy (RM60.09M/yr)

A realized annual visitor spending proxy calculated across 4 covered marine parks (1,464,770 visitors × RM410 avg spend × 10% reef attribution coefficient from Spalding et al. 2017). Covers 32 of 56 units.

Stress Factor Contribution

Calculated via **Tree-Path Decomposition (Saabas 2014)** through the Gradient Boosting ensemble. Decomposes decision paths into 8 feature groups (Substrate, Fish, Impacts, Prior Cover, Geography). **It is NOT NOAA DHW!**

Reef-Adjacent Economy Potential (RM8.7B/yr)

The historical Total Economic Value (TEV) published benchmark by the Department of Marine Park Malaysia (2011–2015 study). Represents total economic capital (primarily non-market existence and bequest value), NOT annual revenue.

Institutional "Green" Status

In ReefSafe, "Green" strictly denotes monitoring units with non-negative predicted annual coral change (≥ 0 pp/yr, Monitor tier): 32 units including **Kapalai (+1.65 pp/yr)**, **Malacca (+1.47 pp/yr)**, and **Redang (+0.65 pp/yr)**. Kapalai is a verified visitor destination.

